

1. —

2. (a) —

$$(b) F_{M_1}(c) = \begin{cases} 0 & \text{se } c < 2 \\ 1/3 & \text{se } 2 \leq c < 3 \\ 1 & \text{se } c \geq 3 \end{cases}, \quad F_{M_2}(c) = \begin{cases} 0 & \text{se } c < 3 \\ 1 & \text{se } c \geq 3 \end{cases}. \quad M_2 \text{ é quase certa.}$$

3. (a) $(\Omega, \mathcal{P}(\Omega), P)$ em que $\Omega = \{(x_1, x_2) : x_i \in \{C_a, C_o\}, i \in \{1, 2\}\}$ e P é a medida de probabilidade de Laplace, i.e.,

$$\begin{aligned} P : \mathcal{P}(\Omega) &\rightarrow [0, 1] \\ A &\rightarrow P(A) = \frac{\#A}{4} \end{aligned}$$

(b) i. X e Y não são iguais.

$$(b) \text{ ii. } F_X(c) = F_Y(c) = \begin{cases} 0 & \text{se } c < 0 \\ 1/4 & \text{se } 0 \leq c < 1 \\ 3/4 & \text{se } 1 \leq c < 2 \\ 1 & \text{se } c \geq 2 \end{cases}; \quad X \text{ e } Y \text{ têm a mesma lei de probabilidade.}$$

4. —

$$5. (a). f_X(a) = \begin{cases} 6/36 & \text{se } a \in \{0, 3\} \\ 10/36 & \text{se } a = 1 \\ 8/36 & \text{se } a = 2 \\ 4/36 & \text{se } a = 4 \\ 2/36 & \text{se } a = 5 \\ 0 & \text{se } c.c. \end{cases}, \quad F_X(c) = \begin{cases} 0 & \text{se } c < 0 \\ 6/36 & \text{se } 0 \leq c < 1 \\ 16/36 & \text{se } 1 \leq c < 2 \\ 24/36 & \text{se } 2 \leq c < 3 \\ 30/36 & \text{se } 3 \leq c < 4 \\ 34/36 & \text{se } 4 \leq c < 5 \\ 1 & \text{se } c \geq 5 \end{cases}$$

$$(b) f_Y(a) = \begin{cases} 1/36 & \text{se } a = 1 \\ 3/36 & \text{se } a = 2 \\ 5/36 & \text{se } a = 3 \\ 7/36 & \text{se } a = 4 \\ 9/36 & \text{se } a = 5 \\ 11/36 & \text{se } a = 6 \\ 0 & \text{se } c.c. \end{cases}, \quad F_Y(c) = \begin{cases} 0 & \text{se } c < 1 \\ 1/36 & \text{se } 1 \leq c < 2 \\ 4/36 & \text{se } 2 \leq c < 3 \\ 9/36 & \text{se } 3 \leq c < 4 \\ 16/36 & \text{se } 4 \leq c < 5 \\ 25/36 & \text{se } 5 \leq c < 6 \\ 1 & \text{se } c \geq 6 \end{cases}$$

$$(c) f_Z(a) = \begin{cases} 11/36 & \text{se } a = 1 \\ 9/36 & \text{se } a = 2 \\ 7/36 & \text{se } a = 3 \\ 5/36 & \text{se } a = 4 \\ 3/36 & \text{se } a = 5 \\ 1/36 & \text{se } a = 6 \\ 0 & \text{se } c.c. \end{cases}, \quad F_Z(c) = \begin{cases} 0 & \text{se } c < 1 \\ 11/36 & \text{se } 1 \leq c < 2 \\ 20/36 & \text{se } 2 \leq c < 3 \\ 27/36 & \text{se } 3 \leq c < 4 \\ 32/36 & \text{se } 4 \leq c < 5 \\ 35/36 & \text{se } 5 \leq c < 6 \\ 1 & \text{se } c \geq 6 \end{cases}$$

$$(d) f_W(a) = \begin{cases} 1/4 & \text{se } a \in \{0, 2\} \\ 1/2 & \text{se } a = 1 \\ 0 & \text{se } c.c. \end{cases}, \quad F_W(c) = \begin{cases} 0 & \text{se } c < 0 \\ 1/4 & \text{se } 0 \leq c < 1 \\ 3/4 & \text{se } 1 \leq c < 2 \\ 1 & \text{se } c \geq 2 \end{cases}$$

(e) Igual à alínea (d)

$$(f) \quad f_S(a) = \begin{cases} 1/36 & se & a \in \{2, 12\} \\ 2/36 & se & a \in \{3, 11\} \\ 3/36 & se & a \in \{4, 10\} \\ 4/36 & se & a \in \{5, 9\} \\ 5/36 & se & a \in \{6, 8\} \\ 6/36 & se & a = 7 \\ 0 & se & c.c. \end{cases}, \quad F_S(c) = \begin{cases} 0 & se & c < 2 \\ 1/36 & se & 2 \leq c < 3 \\ 3/36 & se & 3 \leq c < 4 \\ 6/36 & se & 4 \leq c < 5 \\ 10/36 & se & 5 \leq c < 6 \\ 15/36 & se & 6 \leq c < 7 \\ 21/36 & se & 7 \leq c < 8 \\ 26/36 & se & 8 \leq c < 9 \\ 30/36 & se & 9 \leq c < 10 \\ 33/36 & se & 10 \leq c < 11 \\ 35/36 & se & 11 \leq c < 12 \\ 1 & se & c \geq 12 \end{cases}$$

6. (a) i. $\binom{20}{7}(\frac{1}{2})^{20}$; ii. $\sum_{k=0}^9 \binom{20}{k}(\frac{1}{2})^{20}$; iii. $\sum_{k=15}^{20} \binom{20}{k}(\frac{1}{2})^{20}$; iv. $\sum_{k=4}^{15} \binom{20}{k}(\frac{1}{2})^{20}$

No R: i. `dbinom(7, 20, 1/2)`; ii. `pbinom(9, 20, 1/2)`; iii. `1-pbinom(14, 20, 1/2)`;
iv. `pbinom(15, 20, 1/2)-pbinom(3, 20, 1/2)`

(b) $\binom{6}{6}(\frac{5}{13})^6$; $\binom{6}{0}(\frac{8}{13})^6$

No R: `dbinom(6, 6, 5/13)`; `dbinom(0, 6, 5/13)`

(c) $\sum_{k=6}^{10} \binom{10}{k}(0.2)^k(0.8)^{10-k}$; $\sum_{k=0}^5 \binom{10}{k}(0.2)^k(0.8)^{10-k}$

No R: `1-pbinom(5, 10, 0.2)`; `pbinom(5, 10, 0.2)`

7. (a) $\frac{2}{3}$

(b) i. $\binom{10}{1}\frac{2}{3}(\frac{1}{3})^9$; ii. $\sum_{k=0}^7 \binom{10}{k}(\frac{2}{3})^k(\frac{1}{3})^{10-k}$; iii. $\sum_{k=3}^7 \binom{10}{k}(\frac{2}{3})^k(\frac{1}{3})^{10-k}$; iv. $\frac{\sum_{k=1}^9 \binom{10}{k}(\frac{2}{3})^k(\frac{1}{3})^{10-k}}{\sum_{k=0}^9 \binom{10}{k}(\frac{2}{3})^k(\frac{1}{3})^{10-k}}$

No R: i. `dbinom(1, 10, 2/3)`; ii. `pbinom(7, 10, 2/3)`;
iii. `pbinom(7, 10, 2/3)-pbinom(2, 10, 2/3)`;
iv. `(pbinom(9, 10, 2/3)-pbinom(0, 10, 2/3))/pbinom(9, 10, 2/3)`

(c) $\frac{\binom{10}{1}(1/2)^{10}(4/5)}{\binom{10}{1}(1/2)^{10}(4/5) + \binom{10}{1}(1/3)(2/3)^9(1/5)}$

No R: `dbinom(1,10,1/2)*4/5 / ( dbinom(1,10,1/2)*4/5 + dbinom(1,10,1/3)*1/5 )`

8. (a) $\binom{6}{3}(\frac{1}{2})^6$

No R: `dbinom(3, 6, 1/2)`

(b) $\binom{6}{2}(\frac{1}{10})^2(\frac{9}{10})^4$; $\sum_{k=0}^2 \binom{6}{k}(\frac{1}{10})^k(\frac{9}{10})^{6-k}$; $\sum_{k=2}^6 \binom{6}{k}(\frac{1}{10})^k(\frac{9}{10})^{6-k}$

No R: `dbinom(2, 6, 1/10)`; `pbinom(2, 6, 1/10)`; `1-pbinom(1, 6, 1/10)`

9. (a) $a = 0, b = \frac{3}{4}, d = 1, f(a) = \begin{cases} 1/2 & se & a = 0 \\ 1/4 & se & a \in \{1, 2\} \\ 0 & se & c.c. \end{cases}$

(b) i. $\frac{1}{2}$; ii. $\frac{1}{2}$; iii. $\binom{6}{3}(\frac{1}{4})^3(\frac{3}{4})^3$ **No R:** `dbinom(3, 6, 1/4)`